Tai, Wei Hsuan's Curriculum Vitae

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Education

National Taiwan University Current
B.S. in Biomedical Engineering

Research Interests

Machine Learning, Deep Learning, Computer Vision, Medical Image Analysis

Technical Skills

Programming: C++ (Codeforces current rating: 1400), Python

Machine Learning: PyTorch, Diffusion Models, Self-Supervised Learning

Database: Schema Design, SQL, PostgreSQL, Redis

Web Development: NestJS, Vue, JavaScript, TypeScript

Tools: Linux(Ubuntu), Docker, Git

Research & Project Experience

A multi-task masked autoencoder with GAN-based augmentation for PD-L1 prediction from chest CT images 2026 Spring

- https://www.nature.com/articles/s41598-026-49064-3
- This is a research project that I participated in during 2025 summer. The goal of this project is to utilize the transfer learning method (multi-task masked autoencoder+ViT) to enhance the performance of PD-L1 prediction from chest CT images. We also utilize the GAN-based augmentation method to generate more training data for the multi-task masked autoencoder. The experimental results show that our proposed method can achieve better performance than the baseline methods.
- In this project, I was mainly responsible for the later experiments, provding AUC results for the proposed method. I also participated in the rebuttal process, where I provided the ROC curve, AUPR curve, and the Wilcoxon signed-rank test.

Database Management System Final Project - Uniconnect 2025 Fall

- https://github.com/W-X-Dai/DBMS_FinalProj
- This is the final project for the Database Management System course, where we developed a database management system called Uniconnect. The purpose of Uniconnect is to manage the resources such as internships or gpu servers provided by the university or the enterprises. Enterprises can also find students with specific skills through Uniconnect. In short, this is a platform that connects students, schools and enterprises together.
- I was mainly responsible for the backend development of Uniconnect, which is implemented in NestJS. The database is implemented in PostgreSQL, and I also use Redis for caching. The frontend is implemented in Vue by my teammates, and I utilize Docker to deploy the application. Besides backend, I also contributed to some parts of the frontend and all parts of api communication. (This project is co-worked with LLMs)

Text Recovery with Denoisy Unet 2025 Fall

- https://github.com/W-X-Dai/text_recovery
- This is originally a homework of *Fundamentals of Biomedical Image Processing* course, it originally expected us to utilize CLAHE to make the text more clear. But I think the performance of CLAHE is not good enough, so I implemented a Denoisy Unet to recover the text. The Denoisy Unet is trained on the synthetic dataset generated by adding noise to the original images. The Denoisy Unet can effectively recover the text from the noisy images, and it can also generalize well to the real-world noisy images.

Fundamentals of Biomedical Image Processing Final Project 2025 Fall

- https://bit.ly/4lwuwQy
- This is the final project for the Fundamentals of Biomedical Image Processing course, where we developed a deep learning model to segment the brain tumor from the MRI images with less computational cost. We utilized the BraTS dataset for training and evaluation.
- In this work, I was mainly responsible for the distillation methods. I distilled from a well-trained nnUNet weight to a 2.5D MobileNet based Unet. The distilled model can achieve comparable performance to the original nnUNet, while it is much smaller and faster.